

Ryan Laxton

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EDUCATION

3.67 OVERALL GPA

Stanford University

Palo Alto, CA

Bachelor of Science in Mechanical Engineering

Expected June 2027

- **Relevant Coursework:** Dynamic Systems and Controls, Mechanical Systems Design;
- **Planned:** Electrical Circuits and Electronics, Mechatronic Design

ENGINEERING EXPERIENCE

Test Engineering Lead

Feb. 2025 – Present

Cardinal Propulsion Laboratory, Stanford University

Palo Alto, CA

- Led development and operation of DAQ, instrumentation, and control systems for **3 liquid rocket engines** across **6 hot-fire tests**, supporting engine development from a **50-lbf prototype** to **349-lbf peak thrust**.
- Designed electrical ground support equipment (eGSE) **system architecture, wiring diagrams, and harnessing**, integrating signal conditioning, power distribution, ignition, safety interlocks, and Ethernet-based data acquisition for remote hot-fire operations.
- Developed and validated a **16-channel instrumentation system** for pressure, temperature, and thrust measurement and controls for **6 electromechanical valves**, enabling real-time monitoring and actuation during hot-fire testing.

Mechanical Engineering Intern

June 2026 – August 2026

SkyCool Systems Inc.

Mountain View, CA

- Designed and prototyped a **UV- and corrosion-resistant polymer spring clip** to secure protective jacketing to rooftop PVC piping, optimizing durability, retention, installation force, and manufacturability.
- Developed a **MATLAB parametric model** to optimize clip geometry against bending and yielding; validated designs using **FEA** and iterative **3D-printed prototypes**.
- Diagnosed and repaired an **electromechanical test system**, tracing internal electronics to a failed humidity sensor; prepared drawings and **RFQs** for waterjet and laser-cut components.

Undergraduate Research Assistant

June 2025 – December 2025

Laboratory of Fluids in Complex Environments, Stanford University

Palo Alto, CA

- Executed **300+** controlled high-temperature experiments investigating jet fuel interactions with Leidenfrost surfaces, ensuring repeatable test conditions and consistent data capture.
- Automated droplet size analysis in MATLAB, correlating Sauter Mean Diameter with Weber, Ohnesorge, and thermal ratios to study fuel atomization.

PROJECTS

Motor-Driven Impulse Mechanism | *ME 104: Mechanical Systems Design*

- Designed, fabricated, and experimentally validated a **153-g motor-driven mechanism** generating **1,214 N·s impulse** and **125 N mean force** over **50 mm**.
- Characterized a brushed DC motor and **400:1 planetary gearbox** to select a **2.5-V operating point**, incorporating measured gearbox efficiency and friction losses.
- Performed **stress, buckling, and FEA analyses** to optimize structural geometry and material selection for minimum mass.

Miniature Spark-Gap Tesla Coil | *Personal Project*

- Designed and built a compact spark-gap Tesla coil using a modified consumer high-voltage generator, custom-wound coils, **3 × 3000 V-rated** capacitor bank, and adjustable spark gap; tuned resonance and thermal performance to achieve stable high-frequency discharge and wireless power transfer to fluorescent bulbs.

SKILLS & LEADERSHIP

Technical Skills: MATLAB, Python, SolidWorks, Fusion 360, FEA, Instrumentation & Sensors, NI DAQ, Experimental Testing, 3D Printing, Machining

Instructional Experience: Served as a Supplemental Instruction Leader (Springfield Technical Community College) for two years, assisting over **200** students with Mechanics and E&M coursework.